

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{18} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,0}[m_2] + \frac{1}{12} g_2^4 (g_1^2 + g_2^2) \text{LF}_{4,-1}[m_2] - \frac{4}{45} g_2^4 (g_1^2 + g_2^2) \text{LF}_{5,-2}[m_2] + \right. \\
& \frac{1}{216} \sum_{\mathbf{p}} g_1^6 \text{LF}_{3,0}[m_d^{\mathbf{p}}] + \frac{1}{24} \sum_{\mathbf{p}} g_1^6 \text{LF}_{3,0}[m_e^{\mathbf{p}}] - \frac{1}{288} \sum_{\mathbf{p}} (3 g_1^6 + 19 g_1^2 g_2^4 + 10 g_2^6) \text{LF}_{3,0}[m_l^{\mathbf{p}}] + \\
& \frac{1}{96} \sum_{\mathbf{p}} g_2^4 (g_1^2 + g_2^2) \text{LF}_{4,-1}[m_l^{\mathbf{p}}] + \frac{1}{45} \sum_{\mathbf{p}} g_2^4 (g_1^2 + g_2^2) \text{LF}_{5,-2}[m_l^{\mathbf{p}}] + \\
& \frac{1}{864} (18 \overline{y_u}^{\text{pr}} y_u^{\text{pr}} (g_1^4 + 30 g_1^2 g_2^2 + 45 g_2^4) + \sum_{\mathbf{p}} (g_1^6 - 63 g_1^2 g_2^4 - 90 g_2^6)) \text{LF}_{3,0}[m_q^{\mathbf{p}}] + \\
& \frac{1}{32} g_2^2 (g_1^2 + g_2^2) (-24 \overline{y_u}^{\text{pr}} y_u^{\text{pr}} + \sum_{\mathbf{p}} g_2^2) \text{LF}_{4,-1}[m_q^{\mathbf{p}}] + \frac{1}{15} \sum_{\mathbf{p}} g_2^4 (g_1^2 + g_2^2) \text{LF}_{5,-2}[m_q^{\mathbf{p}}] - \\
& \frac{1}{27} \sum_{\mathbf{p}} g_1^6 \text{LF}_{3,0}[m_u^{\mathbf{p}}] + \frac{1}{3} g_1^4 \overline{y_u}^{\text{pr}} y_u^{\text{pr}} \text{LF}_{3,0}[m_u^{\mathbf{r}}] + \frac{1}{36} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,0}[\tilde{\mu}] + \\
& \frac{1}{24} g_2^4 (g_1^2 + g_2^2) \text{LF}_{4,-1}[\tilde{\mu}] - \frac{2}{45} g_2^4 (g_1^2 + g_2^2) \text{LF}_{5,-2}[\tilde{\mu}] - \frac{1}{12} g_1^6 \text{LF}_{3,3,-3}[m_1, \tilde{\mu}] - \\
& \frac{1}{4} g_1^6 m_1^2 \text{LF}_{3,3,-2}[m_1, \tilde{\mu}] - \frac{1}{3} g_2^4 (g_1^2 + g_2^2) \text{LF}_{2,1,0}[m_2, \tilde{\mu}] - \frac{17}{24} g_2^4 (g_1^2 + g_2^2) \text{LF}_{2,2,-1}[m_2, \tilde{\mu}] + \\
& \frac{1}{6} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,1,-1}[m_2, \tilde{\mu}] + \frac{7}{6} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,2,-2}[m_2, \tilde{\mu}] - \\
& \frac{3}{4} g_2^4 m_2^2 (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_2, \tilde{\mu}] - \frac{1}{4} g_2^4 (2 g_1^2 + 5 g_2^2) \text{LF}_{3,3,-3}[m_2, \tilde{\mu}] + \\
& \frac{1}{4} g_2^4 m_2^2 (2 g_1^2 + g_2^2) \text{LF}_{3,3,-2}[m_2, \tilde{\mu}] - \frac{1}{2} g_2^4 (g_1^2 + g_2^2) \text{LF}_{4,2,-3}[m_2, \tilde{\mu}] + \\
& \frac{1}{2} g_2^4 m_2^2 (g_1^2 + g_2^2) \text{LF}_{4,2,-2}[m_2, \tilde{\mu}] - \frac{1}{12} g_1^4 \tilde{\mu}^2 \overline{y_d}^{\text{pr}} y_d^{\text{pr}} \text{LF}_{2,2,0}[m_d^{\mathbf{r}}, m_q^{\mathbf{p}}] + \\
& \frac{1}{12} g_1^4 \tilde{\mu}^2 \overline{y_d}^{\text{pr}} y_d^{\text{pr}} \text{LF}_{3,1,0}[m_d^{\mathbf{r}}, m_q^{\mathbf{p}}] - \frac{1}{4} g_1^4 \tilde{\mu}^2 \overline{y_e}^{\text{pr}} y_e^{\text{pr}} \text{LF}_{2,2,0}[m_e^{\mathbf{r}}, m_l^{\mathbf{p}}] + \\
& \frac{1}{4} g_1^4 \tilde{\mu}^2 \overline{y_e}^{\text{pr}} y_e^{\text{pr}} \text{LF}_{3,1,0}[m_e^{\mathbf{r}}, m_l^{\mathbf{p}}] + \frac{1}{4} g_1^4 \tilde{\mu}^2 \overline{y_e}^{\text{pr}} y_e^{\text{pr}} \text{LF}_{3,1,0}[m_l^{\mathbf{p}}, m_e^{\mathbf{r}}] + \\
& \frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y_e}^{\text{pr}} y_e^{\text{pr}} (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_l^{\mathbf{p}}, m_e^{\mathbf{r}}] - \frac{3}{8} g_1^2 \tilde{\mu}^2 \overline{y_e}^{\text{pr}} y_e^{\text{pr}} (g_1^2 + g_2^2) \\
& \text{LF}_{4,1,-1}[m_l^{\mathbf{p}}, m_e^{\mathbf{r}}] + \frac{1}{24} \tilde{\mu}^2 \overline{y_e}^{\text{pr}} y_e^{\text{pr}} (3 g_1^2 - g_2^2) (g_1^2 + g_2^2) \text{LF}_{5,1,-2}[m_l^{\mathbf{p}}, m_e^{\mathbf{r}}] + \\
& \frac{1}{12} g_1^4 \tilde{\mu}^2 \overline{y_d}^{\text{pr}} y_d^{\text{pr}} \text{LF}_{3,1,0}[m_q^{\mathbf{p}}, m_d^{\mathbf{r}}] + \frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y_d}^{\text{pr}} y_d^{\text{pr}} (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_q^{\mathbf{p}}, m_d^{\mathbf{r}}] - \\
& \frac{3}{8} g_1^2 \tilde{\mu}^2 \overline{y_d}^{\text{pr}} y_d^{\text{pr}} (g_1^2 + g_2^2) \text{LF}_{4,1,-1}[m_q^{\mathbf{p}}, m_d^{\mathbf{r}}] + \frac{1}{8} \tilde{\mu}^2 \overline{y_d}^{\text{pr}} y_d^{\text{pr}} (3 g_1^2 - g_2^2) \\
& (g_1^2 + g_2^2) \text{LF}_{5,1,-2}[m_q^{\mathbf{p}}, m_d^{\mathbf{r}}] - \frac{1}{2} \overline{y_u}^{\text{pr}} \overline{y_u}^{\text{st}} y_u^{\text{pt}} y_u^{\text{sr}} (g_1^2 + 3 g_2^2) \text{LF}_{2,1,0}[m_q^{\mathbf{p}}, m_q^{\mathbf{s}}] + \\
& \frac{3}{4} \overline{y_u}^{\text{pr}} \overline{y_u}^{\text{st}} y_u^{\text{pt}} y_u^{\text{sr}} (g_1^2 + g_2^2) \text{LF}_{3,1,-1}[m_q^{\mathbf{p}}, m_q^{\mathbf{s}}] + \frac{1}{48} \overline{a_u}^{\text{pr}} a_u^{\text{pr}} (-7 g_1^4 + 30 g_1^2 g_2^2 + 21 g_2^4) \\
& \text{LF}_{2,2,0}[m_q^{\mathbf{p}}, m_u^{\mathbf{r}}] + \frac{1}{48} \overline{a_u}^{\text{pr}} a_u^{\text{pr}} (g_1^4 + 30 g_1^2 g_2^2 + 45 g_2^4) \text{LF}_{3,1,0}[m_q^{\mathbf{p}}, m_u^{\mathbf{r}}] - \\
& \frac{1}{2} g_2^2 \overline{a_u}^{\text{pr}} a_u^{\text{pr}} (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_q^{\mathbf{p}}, m_u^{\mathbf{r}}] - \frac{3}{4} g_2^2 \overline{a_u}^{\text{pr}} a_u^{\text{pr}} (g_1^2 + g_2^2) \text{LF}_{4,1,-1}[m_q^{\mathbf{p}}, m_u^{\mathbf{r}}] + \\
& \frac{1}{48} \overline{a_u}^{\text{pr}} a_u^{\text{pr}} (49 g_1^4 + 30 g_1^2 g_2^2 - 3 g_2^4) \text{LF}_{3,1,0}[m_u^{\mathbf{r}}, m_q^{\mathbf{p}}] + \\
& \frac{1}{16} \overline{a_u}^{\text{pr}} a_u^{\text{pr}} (g_1^2 - 7 g_2^2) (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_u^{\mathbf{r}}, m_q^{\mathbf{p}}] - \\
& \frac{3}{16} \overline{a_u}^{\text{pr}} a_u^{\text{pr}} (7 g_1^2 - g_2^2) (g_1^2 + g_2^2) \text{LF}_{4,1,-1}[m_u^{\mathbf{r}}, m_q^{\mathbf{p}}] + \\
& \frac{1}{8} \overline{a_u}^{\text{pr}} a_u^{\text{pr}} (3 g_1^2 - g_2^2) (g_1^2 + g_2^2) \text{LF}_{5,1,-2}[m_u^{\mathbf{r}}, m_q^{\mathbf{p}}] - \\
& g_1^2 \overline{y_u}^{\text{pr}} \overline{y_u}^{\text{st}} y_u^{\text{pt}} y_u^{\text{sr}} \text{LF}_{2,1,0}[m_u^{\mathbf{r}}, m_u^{\mathbf{t}}] - \frac{1}{8} g_1^2 \text{LF}_{3,1,-1}[\tilde{\mu}, m_1] (g_1^2 + g_2^2)^2 + \\
& \frac{3}{16} g_1^4 (g_1^2 + g_2^2) \text{LF}_{3,2,-2}[\tilde{\mu}, m_1] + \frac{3}{16} g_1^4 m_1^2 (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] + \\
& \frac{1}{24} g_1^2 (g_1^2 + g_2^2) (6 g_1^2 + g_2^2) \text{LF}_{4,1,-2}[\tilde{\mu}, m_1] - \frac{1}{8} g_1^4 (g_1^2 + g_2^2) \text{LF}_{4,2,-3}[\tilde{\mu}, m_1] - \\
& \frac{1}{8} g_1^4 m_1^2 (g_1^2 + g_2^2) \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{24} (-3 g_1^6 - 2 g_1^4 g_2^2 + g_1^2 g_2^4) \text{LF}_{5,1,-3}[\tilde{\mu}, m_1] + \\
& \frac{1}{24} (-9 g_1^4 g_2^2 - 2 g_1^2 g_2^4 + 7 g_2^6) \text{LF}_{3,1,-1}[\tilde{\mu}, m_2] + \\
& \frac{85}{48} g_2^4 (g_1^2 + g_2^2) \text{LF}_{3,2,-2}[\tilde{\mu}, m_2] + \frac{3}{16} g_2^4 m_2^2 (g_1^2 + g_2^2) \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] + \\
& \frac{3}{8} (2 g_1^4 g_2^2 + g_1^2 g_2^4 - g_2^6) \text{LF}_{4,1,-2}[\tilde{\mu}, m_2] - \frac{7}{8} g_2^4 (g_1^2 + g_2^2) \text{LF}_{4,2,-3}$$